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Low luminance deficit and night vision symptoms in intermediate age-related macular degeneration

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ABSTRACT

Background/aims To determine the relationship between self-reported visual difficulties under low luminance conditions (night vision symptoms) and visual function measures in intermediate age-related macular degeneration (AMD).

Methods One hundred participants with bilateral intermediate AMD were examined in a prospective cross-sectional study with visual function measures including best-corrected visual acuity (BCVA), low luminance visual acuity (LLVA) and microperimetry in both eyes. A 10-item Night Vision Questionnaire (NVQ-10) was then used to determine the degree of self-reported night vision symptoms experienced by each participant. For analyses, low luminance deficit (LLD) was derived as the difference between LLVA and BCVA, and microperimetric mean sensitivity (MS; all points) and central sensitivity (CS; points within the central 1°) were determined. Rasch analysis was used to estimate the person measure of night vision symptoms, and its relationship with visual function parameters was determined.

Results NVQ-10 person measures were significantly associated with LLD (β coefficient=0.067, 95% CI 0.005 to 0.130, p=0.034), but not BCVA, LLVA, microperimetric MS or CS (p≥0.090). Participants with the highest degree of self-reported night vision symptoms (fourth quartile of person measure) had significantly worse LLD than those with the least difficulty (first quartile of person measure; p=0.019).

Conclusions In individuals with bilateral intermediate AMD, LLD was associated with self-reported night vision symptoms, suggesting that this measure may better capture the visual difficulties experienced by these individuals under low luminance conditions than the conventional measure of photopic visual acuity.

INTRODUCTION

The early stages of age-related macular degeneration (AMD) are associated with a statistically but not clinically significant reduction in visual acuity measured under optimal illumination and high-contrast conditions.[1 2] However, previous studies have reported that individuals with the early stages of AMD report vision-related difficulties in performing daily activities compared with those with normal retinal health.[3–5] These difficulties are especially marked for activities performed under low luminance conditions.[5–7]

The degree of self-reported visual difficulties under low luminance conditions (referred to as night vision symptoms) has been reported to be associated with an increased risk of developing advanced AMD,[8] especially geographic atrophy (GA).[8 9] In these studies, the degree of night vision symptoms were measured using the 10-item Night Vision Questionnaire (NVQ-10). However, the relationship between the degree of night vision symptoms and clinical measures of visual function under low luminance has not yet been examined. These relationships are important to determine in order to identify clinical measures that can better reflect such visual difficulties, which are not currently captured using the conventional measure of photopic visual acuity.[10]

Specifically, we sought to examine the relationship between night vision symptoms measured by the NVQ-10 and the visual function measures of low luminance visual acuity (LLVA) and microperimetry in this study. Microperimetry was examined since we had previously observed that perimetric retinal sensitivity was reduced and exhibited a greater rate of decline in areas that subsequently developed GA,[11] and LLVA was examined since it exhibits a close relationship with microperimetric retinal sensitivity.[2] In addition, these two measures are performed under low luminance conditions, which may better reflect the degree of visual difficulty under low luminance conditions assessed by the NVQ-10 than the conventional measure of photopic visual acuity.

METHODS

This study involved the prospective recruitment of a consecutive series of participants with bilateral intermediate AMD, and it was approved by the Human Ethics Committee of the Royal Victorian Eye and Ear Hospital (RVEEH) and was conducted in accordance with the Declaration of Helsinki. Written informed consent was obtained from all participants.

Participants

Participants were recruited from private ophthalmology clinics and the medical retina clinics at the RVEEH. The inclusion criteria were being 50 years of age or older, having a best-corrected visual acuity (BCVA) of 20/40 (or 0.30 logarithm of the minimum angle of resolution (logMAR)) or better and having intermediate AMD (defined as having at least drusen >125 µm[12]) in both eyes. The exclusion criteria included the presence of any choroidal neovascularisation, any atrophic changes including GA detected on colour fundus photography, drusen-associated atrophy or nascent GA detected on spectral-domain optical coherence tomography (SD-OCT),[13] significant cataracts, glaucoma, amblyopia or any corneal pathology that would compromise vision in either eye. Participants were also excluded if they had diabetes, had any neurologic or systemic disease affecting vision, were taking any medication known to affect retinal

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function, had any physical or mental impairment preventing them from participating in this study or were unable to sign a consent form.

Examination procedures

Visual acuity measurements and microperimetry examinations were performed prior to fundus imaging, to prevent bleaching of the retina from the imaging light sources.

Visual acuity measurements

A standardised refraction protocol was performed monocularly using an Early Treatment of Diabetic Retinopathy Study (ETDRS) refraction chart at a testing distance of 4 m and with a standard test luminance of 85 cd/m^2. BCVA was also measured at the same distance. A 2.0-log unit neutral density filter (Kodak Wratten filter; Kodak, Rochester, New York, USA) was then placed in front of the study eye and LLVA was measured without additional time for adaptation, since a previous study has shown that no additional period of adaptation was required for eyes with GA when using this procedure to measure LLVA.[14 15] For all measurements, participants were asked to guess all the letters in a given line, and visual acuity measurements were recorded in logMAR, with each letter given a score of 0.02 logMAR.

Microperimetry examination

Microperimetric examinations were performed using the Macular Integrity Assessment (MAIA, CenterVue, Padova, Italy) microperimeter as outlined in detail previously.[16] Briefly, they were performed following pupillary dilation using 1 drop of 1% tropicamide and 1 drop of 2.5% phenylephrine to achieve a pupil diameter of at least 6 mm. All participants were given identical verbal instructions regarding how to perform the microperimetry examination.

The MAIA performs fundus tracking using a line-scanning laser ophthalmoscope with a super-luminescent diode illumination that has a central wavelength of 850 nm. It uses the entire fundus as a reference when performing fundus tracking, capturing the fundus images at 25 frames per second. With fundus tracking, test stimuli can be presented at the same retinal location throughout the test independent of shifts in fixation. A red circular fixation target of 1° diameter was used in this study, and Goldman III stimuli were presented for 200 ms against a background of 1.27 cd/m^2 using a 4–2 threshold strategy. The maximum stimulus luminance was 318 cd/m^2, creating a dynamic range of 36 dB of differential contrast. A customised stimulus grid (CERA AMD 6° grid) consisting of 37 points located at 0°, 1°, 2.33°, 4° and 6° from fixation was used in this study, designed by our group specifically for the assessment of the macular region in AMD.

In this study, two examinations were performed on the right eye followed by one examination on the left eye for all participants; approximately 3 min of rest was given between each test to minimise the effect of fatigue. The first examination of the right eye was discarded from the analyses to avoid the influence of an intrasession learning effect, as previously reported.[16] Test reliability for the second examination was assessed by the frequency of false-positive responses, measured by presentations of suprathreshold stimulus to the physiological blind spot, which was manually located on the MAIA before the presentation of the first stimuli. Any test with false-positive responses of >25% was considered unreliable and was repeated. This cut-off was chosen because the MAIA presents a false-positive stimulus approximately every 1 min, and there are typically only four to five false-positive stimuli presented in each test given the short duration of the examinations.

The average retinal sensitivity of all points within the stimulus grid in this study was termed mean sensitivity (MS). The average of the five stimuli within the central 1° (comprising of one point at 0° and four points at 1°) was termed central sensitivity (CS), and was analysed in this study since it reflects foveal function, and we have found previously that it corresponds closely with visual acuity measurements.[2]

Imaging and fundus grading

Fundus imaging was performed using colour fundus photography (Canon CR6-45NM; Canon, Saitama, Japan), and near-infrared reflectance (NIR), fundus autofluorescence (FAF) and SD-OCT line and volume scans all performed on a Spectralis HRA+OCT (Heidelberg Engineering, Heidelberg, Germany). A 45° diameter colour fundus photograph and 30°×30° NIR and FAF images centred on the macular were acquired. For SD-OCT, a high-resolution horizontal single-line scan was performed with 100 frames averaged, followed by a volume scan of 49 B-scans of the central 20°×20° area, with 25 frames averaged for each B-scan. An experienced grader performed all the grading on colour fundus photographs[17] to determine the presence of intermediate AMD according to a recent clinical classification system.[12] The multimodal imaging modalities were then used to ensure that the inclusion and exclusion criteria were met (such as the absence of atrophic changes).

Night Vision Questionnaire

The NVQ measures night vision functioning using a 10-item scale derived from four items of the National Eye Institute Vision Functioning Questionnaire (NEI-VFQ) 25 items and six items of a night vision symptom list.[8] Response options are Likert scales with five options rating difficulty ('None', 'A little difficulty', 'Moderate difficulty', 'Extreme difficulty', 'Stopped doing this because of your eyesight'; item 1–4) or four options rating impact ('Not at all bothered', 'A little bothered', 'Somewhat bothered', 'Very bothered'; item 5–10). Items 1–3 could not be scored if the participant answered 'Stopped doing this for other reasons or not interested in doing this' or 'Not currently driving'. Given the ubiquity of lighting at night-time in the urbanised society where this cohort was examined, the item content is related to mesopic rather than scotopic visual function. Thus, we refer to the construct captured by the NVQ-10 as *night vision symptoms*, but we assessed its association with measures of mesopic rather than scotopic visual function. A list of all items can be found in online supplementary table S1.

Statistical analysis

We performed Rasch analysis on the NVQ-10 using Winsteps software (V.3.68, Chicago, Illinois, USA) to estimate scores that approximate interval-level measurement (person measure; expressed in log of the odds units (logits)) from raw ordinal data. The NVQ-10 person measures capture night vision symptoms, with a higher person measures score indicating more severe night vision symptoms.

In this study, the eye with better BCVA was specified as the study eye for all visual function measures to be analysed. When BCVA was the same in both eyes, the eye with the better LLVA was specified as the study eye; the study eye was randomly allocated using computer-generated numbers if LLVA was also the same in both eyes. The association between the person measure score of night vision symptom estimated from Rasch analysis

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Table 1 Characteristics of visual function measures from the better eye of the study participants

Parameter		Mean±SD	Range
Visual acuity (logMAR)	Best-corrected visual acuity	−0.03±0.09	−0.22 to 0.16
	Low luminance visual acuity	0.24±0.15	−0.02 to 0.80
	Low luminance deficit	0.27±0.12	0.10 to 0.80
Microperimetry (dB)	Mean sensitivity	26.9±1.7	21.0 to 30.8
	Central sensitivity	26.5±2.1	20.0 to 30.0

logMAR, logarithm of the minimum angle of resolution.

and clinical parameters (including age and the visual function measures) were determined using univariate linear regression analyses. The person measure score was then grouped into quartiles, and the visual function measures were compared between the quartiles using a Kruskal–Wallis test with post hoc Bonferroni tests. All statistical analyses were performed using commercially available statistical software (IBM SPSS Statistics, software V.21; IBM/SPSS, Chicago, Illinois, USA).

RESULTS

A total of 100 participants were included in this study. Their mean age was 69.8±7.4 years (range 51–81 years), 74 participants (74%) were female and 13 participants (13%) were not currently drivers (having answered 'Stopped doing this for other reasons or not interested in doing this' or 'Not currently driving' in response to items 1–3). The characteristics of the visual function measures are outlined in table 1.

Night vision symptoms and visual function measures

The mean Rasch-generated person measure score for self-reported night vision symptoms for this cohort was −2.85 ±1.86 logits (range −5.78 to 1.50). Using univariate linear regression analyses, the person measure score was only significantly associated with low luminance deficit (LLD; β coefficient=0.067, 95% CI 0.005 to 0.130, p=0.034), but not age, BCVA, LLVA and microperimetric MS and CS (p≥0.090; table 2). These findings remained the same after including age as a covariate in each of the univariate linear regression analyses (data not shown).

Grouping participants into quartiles based on the Rasch-generated person measure score for self-reported night vision symptoms, there were no significant differences in BCVA, LLVA and microperimetric MS and CS between these groups (p≥0.098). However, there was a significant difference between the groups for LLD (p=0.012), and post hoc analysis revealed that LLD was significantly worse for participants in the fourth quartile (median, IQR=15.5 letters, 12–19 letters) compared with participants in the first quartile (median, IQR=12.0 letters, 9–15 letters; p=0.019), but not the second and third quartile (p=1.000).

Table 2 Factors associated with night vision functioning on univariate linear regression analysis

Parameter	β Coefficient	95% CI	p Value
Age (years)	0.031	−0.019 to 0.091	0.223
Visual acuity (letters)*			
Best-corrected visual acuity (BCVA)	−0.001	−0.086 to 0.084	0.984
Low luminance visual acuity (LLVA)	−0.043	−0.093 to 0.007	0.090
Low luminance deficit (LLD)	0.067	0.005 to 0.130	0.034
Microperimetry (dB)*			
Mean sensitivity (MS)	−0.145	−0.366 to 0.077	0.197
Central sensitivity (CS)	−0.107	−0.280 to 0.066	0.222

* Visual acuity and microperimetry parameters are obtained from the better eye.

DISCUSSION

In this study, LLD but not microperimetric retinal sensitivity was significantly associated with the degree of self-reported night vision symptoms measured using the NVQ-10 in individuals with bilateral intermediate AMD. Since the degree of these symptoms have been previously reported to confer an increased risk of progression to advanced AMD,[8 9] LLD may also potentially confer such prognostic information and can be used as a clinical visual function parameter to reflect the degree of such symptoms. Therefore, the findings of this study suggest that LLD may be a useful visual function parameter that can be easily implemented in clinical settings and may better capture the degree of night vision symptoms experienced by individuals with intermediate AMD than the conventional measure of BCVA.

To date, this is the first study to examine the relationship between the degree of self-reported night vision symptoms measured using the NVQ-10 and clinical measures of visual function under low luminance conditions. The only other questionnaire designed specifically to assess self-reported low luminance visual difficulties is the Low Luminance Questionnaire (LLQ), containing 32 items which assesses problems with activities under low luminance conditions.[6] Using the LLQ, persons with AMD have been found to report a significant impairment independent of their degree of visual impairment.[18] The self-reported low luminance difficulty as measured by the LLQ has also been found to correlate with the rod-mediated, but not cone-mediated, components of dark adaptation measured at a peripheral retinal location.[6] Since the two studies are not directly comparable because of the measures used and disease severity of the participants included, it remains to be examined whether rod-mediated measures better correlate with self-reported night vision symptoms as measured by the NVQ-10 in eyes with intermediate AMD without substantial loss of BCVA. However, it is more likely that a battery of visual function measures is required to capture the extent of self-reported night vision symptoms, since different activities under low luminance conditions are mediated by different components of vision; for example, 'reading in dim light' (item 6 on the NVQ-10) is likely to require foveal cone-mediated function, as compared with a more general 'vision at night' (item 5 on the NVQ-10) which may involve rod-mediated function predominantly. Given the ubiquity of lighting at nighttime in the urbanised society where this cohort was examined, vision at night is unlikely to be only rod mediated.

In this study, we found that LLD, but not BCVA or LLVA, was associated with the degree of night vision symptoms by the NVQ-10. This is likely due to individuals reporting the degree of visual difficulty under low luminance condition relative to those under optimal lighting conditions as their intraindividual reference, in a similar way that LLD reflects visual function under low luminance conditions relative to those under optimal lighting conditions. We also found in this study that only LLD, but not microperimetric retinal sensitivity, was found to be associated with the degree of night vision symptoms by the NVQ-10, even though we have previously found a significant relationship between central retinal sensitivity and LLVA.[2] Although both measures are performed under a similar luminance level, they capture distinctly different aspects of visual

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function. Microperimetry measures luminance increment thresholds (and thus varying levels of contrast against the background) and involves only the visual process of detection, while the measure of LLVA involves the visual processes of detection, resolution and identification at a single level of contrast.[19] Thus, although microperimetric retinal sensitivity may detect a greater degree of foveal functional deficit than the visual acuity measures,[2] the findings of this study suggest that visual acuity measures under low luminance conditions may better reflect visual difficulties experienced by individuals. These findings are especially important when considering relevant clinical end points—namely, 'a characteristic or variable that reflects how a patient feels, functions or survives', as defined by the Biomarkers Definitions Working Group[20]—for interventional studies targeting the early stages of AMD.[10 21]

There are several limitations in this study that are important to note, including the small sample size and the lack of a control group, with the latter meaning that the degree of self-reported night vision difficulties experienced by healthy individuals and individuals with intermediate AMD could not be directly compared. However, given that the primary aim of this study was to examine the associations between such night vision difficulties and visual function measures in individuals with intermediate AMD, the lack of a control group does not detract from our interpretation of the results. In addition, the highly standardised measurements of visual function in this cohort and use of Rasch analysis provides confidence that these findings are valid for the sample we examined, and can be extrapolated with caution. While the inclusion criteria of having a BCVA of 20/40 or better may also have limited the range of participants examined, these participants were still characterised by having a wide range of different visual functions when measured using LLVA and microperimetry. In addition, the majority of individuals with intermediate AMD do not have a BCVA worse than this cut-off unless they suffer from additional ocular disease, which makes our sample representative of the majority of individuals with intermediate AMD. This criterion was also chosen since it fell within the vision requirements for driving in Australia, and allowed an examination of participants that would be otherwise considered to have relatively preserved vision clinically. Although the degree of night vision symptoms have been found to confer an increased risk of progression to advanced AMD,[8 9] it also remains to be determined in future longitudinal studies whether LLD confers prognostic value for the early stages of AMD, given the association found between these two parameters found in this study. Despite previous studies demonstrating that a period of dark adaptation was not required when measuring LLVA,[14 15] there is a possibility that we might have underestimated LLVA when using this method in this study. However, this procedure was consistent for all the measurements performed in this study, and is therefore unlikely to influence the results of this study. Future studies would also benefit from examining near visual acuity (which was not examined in this study), since reading is an aspect of visual difficulty often reported by participants with the early stages of AMD that can be measured clinically.

In conclusion, this study showed that the degree of self-reported visual difficulty under low luminance conditions in individuals with intermediate AMD is significantly associated with LLD, but not microperimetric retinal sensitivity. These findings suggest that LLD may be a useful visual function measure that can be easily implemented in clinical settings to reflect such visual difficulty, beyond what is captured by the conventional measure of photopic visual acuity.

Contributors All authors were involved in the design and conduct of the study, the collection and management of the data, analysis and interpretation of the data and preparation, review or approval of the manuscript.

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Patient consent Obtained.

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